INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 14

1. Experiment Title:

Creating and Using a View

2. Aim:

Create a view containing students with marks of 70 or more and retrieve rows from that view.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

A view is a named query that can be used like a virtual table. CREATE VIEW stores the query definition. SELECT can retrieve data from the view.

5. Algorithm / Procedure:

1. Create score and insert the four supplied student records.
2. Create the high_scores view with name and marks for marks >= 70.
3. Select all rows from high_scores.
4. Select only names from high_scores and sort them alphabetically.

6. Program / Queries / Commands:

```
CREATE TABLE score (
    name VARCHAR(30),
    subject VARCHAR(20),
    marks INT
);

INSERT INTO score (name, subject, marks)
VALUES
    ('Asha', 'Maths', 78),
    ('Rohan', 'Maths', 64),
    ('Neha', 'Maths', 88),
    ('Kiran', 'Maths', 52);

-- Task 1
CREATE VIEW high_scores AS
SELECT name, marks
FROM score
WHERE marks >= 70;

-- Task 2
SELECT * FROM high_scores;

-- Task 3
SELECT name FROM high_scores
ORDER BY name;
```

7. Input:

Asha—78, Rohan—64, Neha—88, Kiran—52.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 2: name	marks
Asha	78
Neha	88
Task 3: name	
Asha	
Neha	

9. Result:

The SQL statements for Experiment 14 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is a view in SQL?
2. Which command creates a view?
3. Does a view store a query definition?
4. How can rows be retrieved from a view?